REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

sUPI PAYLOAD BOARD

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IDEAL DIODE	5
RAIL DISCHARGE	6
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Block Diagram

File: 100068622_block_diagram.kicad_sch

RELATED PARTS			
TYPE	NUMBER	CFG	REV
PWB	100068610	001	1
PWA	100068622	001	1
EBOM	100068622	001	1

DRAWING MEETS REQUIREMENTS OF MIL-STD-31000

DO NOT SCALE THIS DRAWING

UNLESS OTHERWISE SPECIFIED
CAPACITORS ARE IN MICROFARADS
RESISTORS ARE IN OHMS
DIMENSIONS ARE IN INCHES

TOLERANCE ON
DECIMALS ANGLES
± .01 ± .1°
± .005

CONTRACT NO.		
DWN		
CHK		
APVD		
APVD		
APVD		
APVD		
BY DIRECTION		DATE

PCB SCHEMATIC
SUPI PAYLOAD BOARD

SIZE	CAGE CODE	DRAWING NO.	REV
D		100046137	2
SCALE 1:1	FILENAME: 100068622-001_Rev1.kicad_sch		
	SHEET:		SHEET: 1/7

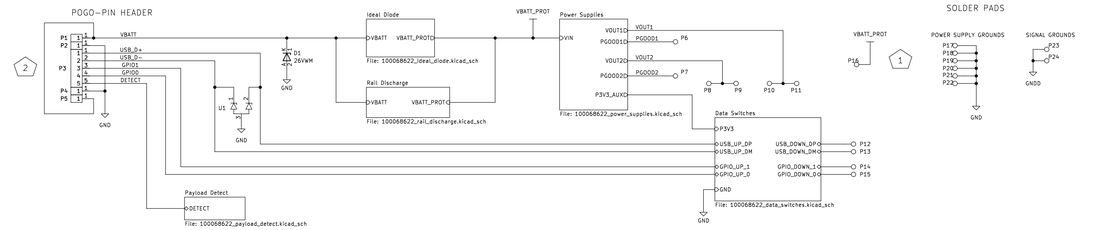
DESIGN NOTES

1. VBATT IS EXPOSED TO PAYLOAD AFTER IDEAL DIODE TO PREVENT ADDITIONAL PAYLOAD CAPACITANCE FROM TRANSFERRING CHARGE BACK TO THE PLATFORM

2. CONNECTOR PINOUT IS MIRRORED FROM PLATFORM

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
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FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
FID5 Fiducial
FID6 Fiducial



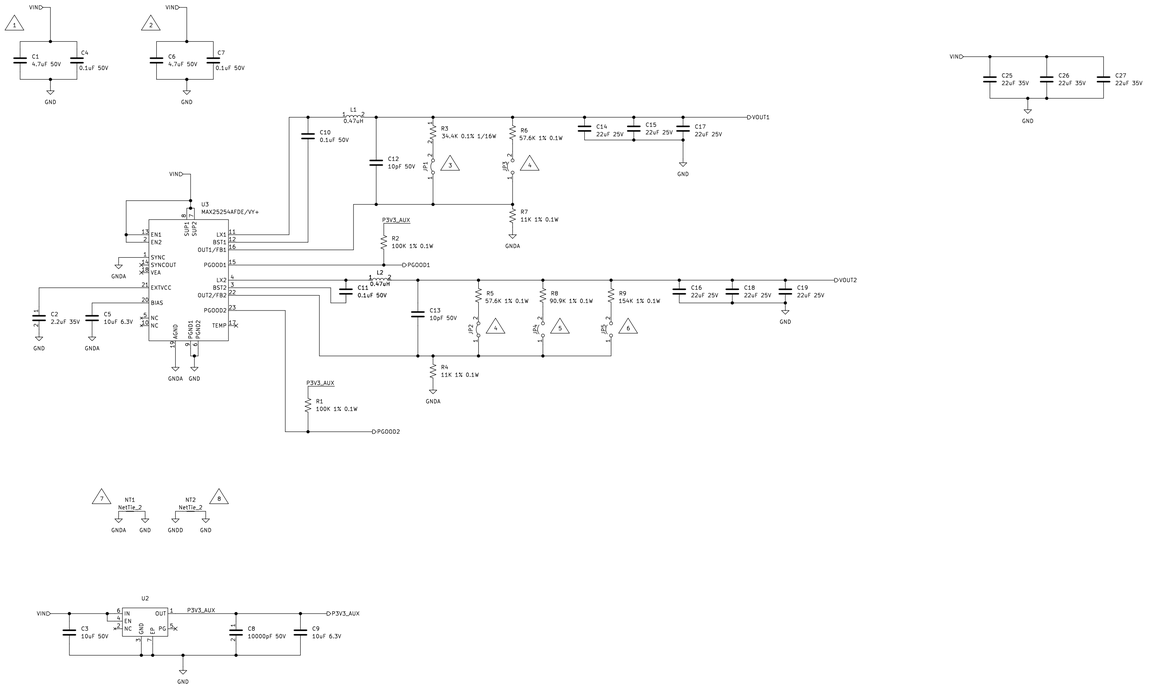
SIZE D	CAGE CODE	DRAWING NO. 100046137	REV 2
SCALE 1:1	FILENAME: 100068622_block_diagram.kicad_sch		
	SHEET: Block Diagram		SHEET: 2/7

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

LAYOUT GUIDANCE

1. PLACE NEAR U1 PINS 8 AND 9
2. PLACE NEAR U1 PINS 6 AND 7
3. MARK SILKSCREEN AS "3V3"
4. MARK SILKSCREEN AS "5V"
5. MARK SILKSCREEN AS "7V4"
6. MARK SILKSCREEN AS "12V"
7. CONNECT GND TO GNDA AT A SINGLE POINT AWAY FROM NOISE
8. CONNECT GND TO GNDD AT A SINGLE POINT AWAY FROM NOISE

POWER SUPPLIES



SIZE D	CAGE CODE	DRAWING NO. 100046137		REV 2
SCALE 1:1	FILENAME: 100068622_power_supplies.kicad_sch			
	SHEET: Power Supplies		SHEET: 3/7	

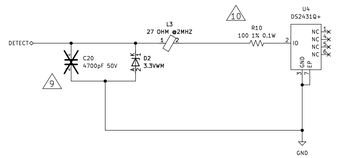
REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

LAYOUT GUIDANCE

9 PLACE C26 AND D1 NEXT TO P12

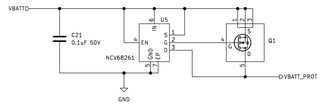
10 PLACE R8 NEXT TO U2

PAYLOAD DETECT



SIZE: D	CAGE CODE:	DRAWING NO.: 100046137	REV: 2
SCALE: 1:1	FILENAME: 100068622_payload_detect.kicad_sch		
	SHEET: Payload Detect		SHEET: 4/7

IDEAL DIODE

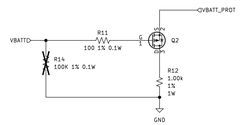


REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

SIZE D	CAGE CODE	DRAWING NO. 100046137	REV 2
SCALE 1:1	FILENAME: 100068622_ideal_diode.kicad_sch		
	SHEET: Ideal Diode		SHEET: 5/7

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

RAIL DISCHARGE



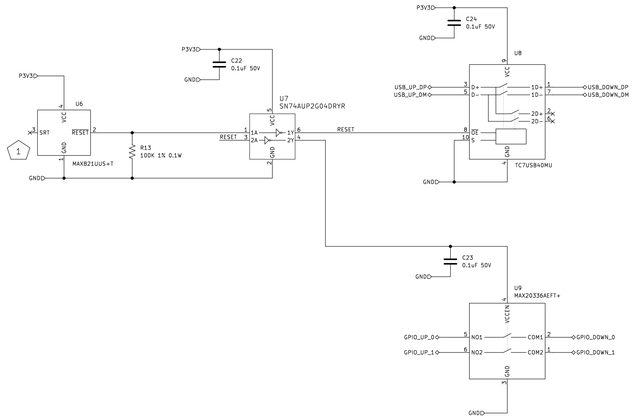
SIZE: D	CAGE CODE:	DRAWING NO.: 100046137	REV: 2
SCALE: 1:1	FILENAME: 100068622_rail_discharge.kicad_sch		
	SHEET: Rail Discharge		SHEET: 6/7

DESIGN NOTES

1. POWER-ON RESET DELAY IS 100MS MINIMUM, 200MS MAXIMUM, 160MS TYPICAL. SWITCHES WILL DISCONNECT AS SOON AS 3.3V AUX RAIL DROPS BELOW THRESHOLD VOLTAGE.

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

DATA SWITCHES



SIZE	CAGE CODE	DRAWING NO.	REV
D		100046137	2
SCALE	FILENAME: 100068622_data_switches.kicad_sch		
1:1	SHEET: Data Switches		SHEET: 7/7